

KITCHEN SECURITY ALARM Using MQ2 Sensor

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The most common cause of fire in a kitchen is gas leakage. This project will help you make an LPG leakage cum smoke alarm for your kitchen using a gas sensor. The sensor will detect the intensity of LPG gas as well as smoke, if any. If the intensity of LPG gas or smoke exceeds a certain limit, an audible alarm will notify to take care of the leakage as soon as possible.

LPG/smoke detectors are amazing devices as they are small and low-cost, yet very useful. MQ2 is the most common metal oxide semiconductor (MOS) type of gas sensor. It is very sensitive to smoke and flammable gases like LPG, butane, propane, methane, alcohol, hydrogen, and carbon monoxide. When a gas comes in contact with the sensor, the circuit uses a simple voltage divider network to detect it.

The MQ2 sensor has two electrodes made of aluminium oxide (Al_2O_3) and a heating element made of tin dioxide (SnO_2) that acts as the main sensing layer.

Circuit and working

Circuit diagram of the alarm using MQ2 sensor is shown in Fig. 1. The circuit is built around step-down transformer X1, bridge rectifier BR1, 5V voltage regulator LM7805 (IC1), MQ2 sensor (Sensor1), op-amp LM358 (IC2), timer NE555 (IC3), 5V SPDT relay RL1, piezo buzzer (BZ1), and a few other components. Capacitor C1 connected across the supply terminals minimises voltage ripples and any noise signals.

The circuit needs 5V DC to operate, which is derived from the 9V, 500mA secondary side of transformer X1. The 230V AC mains power supply is connected to the primary of X1 via connector CON1 in the circuit. The 9V AC secondary of X1 is connected to bridge rectifier BR1 for rectification. The rectified voltage filtered by capacitor C1 is given to IC LM7805 to get regulated 5V for the circuit. LED1 indicates availability of 5V.

The output of MQ2 sensor is given to the non-inverting pin 3 of IC LM358, which acts as a comparator in this circuit. The inverting pin 2 of LM358 is connected to pot VR1 so that sensitivity of the circuit can be adjusted. The output of LM358 is given to the base of buzzer driver transistor T1 via resistor R7. The

PARTS LIST

Semiconductors:

- IC1 - LM7805, 5V voltage regulator
- IC2 - LM358 op-amp
- IC3 - NE555 timer
- T1, T2 - 2N2219 NPN transistor
- BR1 - 1A bridge rectifier
- LED1 - 5mm LED
- D1 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R2, R4, R5 - 1-kilo-ohm
- R3, R7 - 680-ohm
- R6 - 1-mega-ohm

Capacitors:

- C1 - 1000 μ F, 35V electrolytic
- C2 - 1000 μ F, 16V electrolytic
- C3 - 10nF ceramic disk

Miscellaneous:

- Exhaust fan - 230V AC operated
- RL1 - 5V, SPDT relay
- CON1-CON3 - 2-pin connector
- Sensor1 - MQ2 gas sensor
- X1 - 230V AC primary to 9V, 500mA secondary transformer

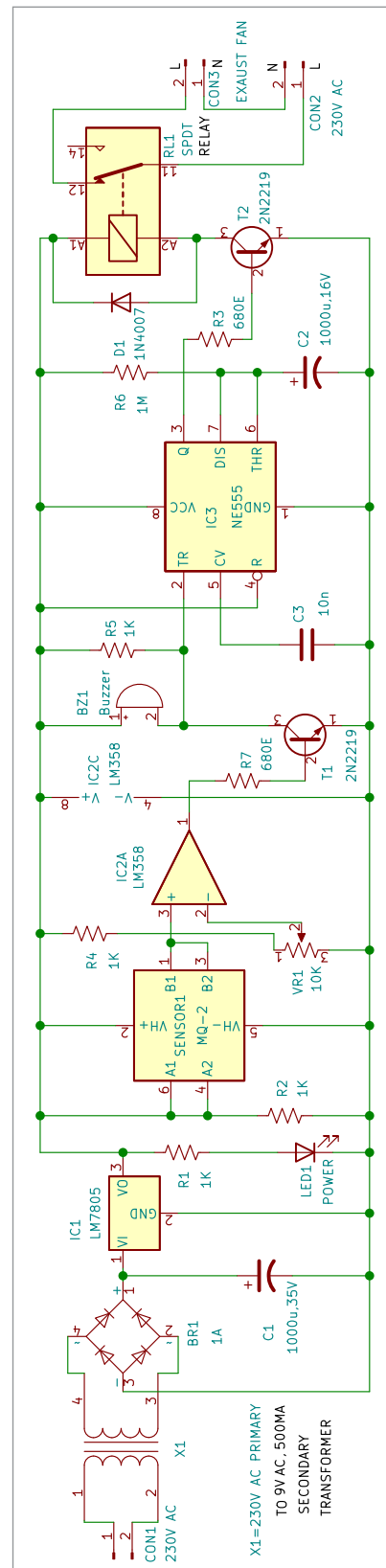


Fig. 1: Circuit diagram